

CHIEVRES AB, Belgium

WMO: 064320

Lat: 50.572N

Lon: 3.831E

Elev: 193

StdP: 14.59

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.0	24.7	14.5	11.5	21.9	19.5	14.7	26.3	31.1	46.5	27.9	47.0	6.7	40	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	85.4	68.2	81.6	66.9	77.8	65.0	70.0	81.5	68.2	78.8	66.5	75.6	7.0	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.1	96.8	74.4	64.5	91.4	72.1	62.9	86.6	70.5	34.0	81.7	32.5	79.1	31.2	75.5	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.2	21.0	18.4	16.9	91.4	6.6	3.1	12.1	93.6	8.3	95.4	4.5	97.2	-0.3	99.4		
(5)			16.2	73.6	6.2	2.2	11.7	75.1	8.1	76.4	4.6	77.6	0.0	79.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	38.3	39.9	44.4	49.2	55.5	61.1	64.7	64.4	58.9	53.0	45.1	39.8	(6)	(7)
	DBStd	11.23	7.54	6.86	5.95	6.22	6.47	5.67	5.49	5.08	4.94	6.18	6.51	7.26	(7)	(8)
	HDD50	1477	365	285	188	88	18	0	0	0	1	40	172	320	(8)	(9)
	HDD65	5185	828	704	638	474	304	144	74	74	195	372	598	780	(9)	(10)
	CDD50	1937	2	1	15	64	187	335	457	446	267	134	24	5	(10)	(11)
	CDD65	171	0	0	0	0	9	29	66	55	11	1	0	0	(11)	(12)
	CDH74	1713	0	0	0	17	107	313	632	543	95	6	0	0	(12)	(13)
	CDH80	478	0	0	0	1	22	81	192	168	14	0	0	0	(13)	(14)
(14) Wind	WSAvg	8.2	10.0	9.9	9.3	8.0	7.6	7.1	6.7	6.7	6.8	7.7	8.8	9.8	(14)	(15)
(15) Precipitation	PrecAvg	30.2	2.4	2.0	2.7	2.0	2.5	2.9	2.7	2.2	2.2	2.9	3.0	2.8	(15)	(16)
	PrecMax	40.8	4.7	4.4	5.1	3.7	4.8	5.7	7.1	4.5	6.7	7.0	6.6	7.0	(16)	(17)
	PrecMin	20.4	0.9	0.3	0.6	0.4	0.5	0.2	0.6	0.2	0.3	0.3	1.1	0.5	(17)	(18)
	PrecStd	5.4	1.1	1.0	1.2	1.0	1.1	1.4	1.6	1.1	1.3	1.7	1.3	1.5	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	57.3	65.6	76.8	83.2	87.6	90.6	89.7	81.9	73.5	62.7	56.2	(19)	(20)
		MCWB	52.0	48.2	53.5	60.9	65.8	68.8	69.3	69.5	66.6	63.2	57.6	52.8	(20)	(21)
	2%	DB	53.1	53.3	60.9	70.9	76.6	81.8	85.2	84.7	76.7	68.4	58.9	54.2	(21)	(22)
		MCWB	50.6	49.1	52.3	57.5	62.1	66.8	68.3	67.7	64.4	61.2	54.8	51.7	(22)	(23)
	5%	DB	50.4	51.2	56.9	64.6	72.5	77.2	81.2	80.3	71.9	64.6	56.4	52.1	(23)	(24)
		MCWB	47.7	48.0	50.4	54.5	60.5	65.2	67.1	66.3	62.4	58.9	53.4	49.7	(24)	(25)
	10%	DB	48.3	49.4	53.8	60.3	67.9	72.8	76.9	75.6	68.3	62.0	54.2	49.9	(25)	(26)
		MCWB	45.9	46.6	49.0	52.3	58.7	62.9	65.1	64.3	60.0	57.7	51.5	47.7	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	51.6	55.4	62.1	68.2	71.3	72.4	72.2	68.7	64.5	58.4	54.0	(27)	(28)
		MCDB	54.5	54.4	61.8	74.1	79.8	82.1	84.3	85.5	79.2	71.1	61.4	55.5	(28)	(29)
	2%	WB	50.8	49.9	53.2	58.3	64.6	68.4	70.1	69.3	66.1	62.0	55.4	52.0	(29)	(30)
		MCDB	52.8	52.0	58.5	67.9	72.6	78.6	82.1	80.4	74.2	67.3	58.1	53.9	(30)	(31)
	5%	WB	48.2	48.4	51.4	55.6	61.6	66.2	68.3	67.4	63.7	60.1	53.8	50.0	(31)	(32)
		MCDB	50.2	50.8	56.1	63.4	69.9	75.4	78.9	77.7	69.2	64.0	56.3	52.0	(32)	(33)
	10%	WB	46.0	46.7	49.5	53.2	59.4	63.9	66.3	65.5	61.7	58.3	51.8	47.9	(33)	(34)
		MCDB	48.0	49.2	53.3	59.2	67.1	71.5	74.7	73.9	66.9	61.5	54.0	49.8	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	10.0	13.3	16.5	17.1	17.2	18.3	18.4	16.5	13.3	9.4	8.1	(35)	(36)
		MCDBR	9.4	12.3	19.4	24.3	24.5	25.2	25.6	26.6	22.5	16.7	10.7	9.0	(36)	(37)
		MCWBR	8.3	9.3	12.4	13.3	12.4	13.0	11.5	11.9	11.6	10.5	8.3	8.0	(37)	(38)
	5% WB	MCDBR	9.7	10.8	16.4	21.7	21.2	21.9	22.8	23.9	18.8	14.8	9.9	8.7	(38)	(39)
		MCWBR	8.8	8.8	11.2	12.3	11.7	11.9	11.0	11.7	10.3	9.8	8.1	8.1	(39)	(40)
(40) Clear-Sky Solar Irradiance	taub		0.328	0.349	0.385	0.417	0.428	0.424	0.426	0.417	0.389	0.366	0.340	0.325	(40)	(41)
	taud		2.387	2.349	2.249	2.179	2.194	2.226	2.225	2.272	2.335	2.397	2.387	2.358	(41)	(42)
	Ebn at Noon		220	246	256	261	263	265	262	258	252	237	213	201	(42)	(43)
	Edh at Noon		20	27	36	42	43	43	42	38	32	25	20	18	(43)	(44)
(44) All-Sky Solar Radiation	RadAvg		256	473	845	1344	1574	1724	1637	1391	1057	621	301	202	(44)	(45)
	RadStd		29	92	115	148	146	157	176	136	90	53	34	29	(45)	(46)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page